



AI-Powered CRM Platform - System Analysis Document

CNC111 - Web Programming Project



Team Members

Name	Student ID
Moaz Hassan Ismail El Garawany	320240032
Ahmed Hossam Aldeen Alsayed	320240051
Ahmed Mahmoud	320240024



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System Overview

The **AI-Powered CRM Platform** is a modern Customer Relationship Management system that leverages artificial intelligence to enhance lead management, appointment scheduling, and customer interactions. The system is built using Next.js 14+, Supabase, and integrates with OpenAI GPT-4o for intelligent automation.

Key Modules:

- **LeadCatch Module** - AI-powered lead capture, parsing, and enrichment
- **Appointments Module** - Smart scheduling with calendar integrations
- **CRM Core** - Contact management, sales pipeline, and analytics
- **AI Module** - Shared AI services for all modules



Use Case Diagram

```
flowchart TB
    subgraph Actors
        U[👤 User/Sales Rep]
        A[👤 Admin]
        AI[🤖 AI System]
        CS[📅 Calendar Service]
        ES[🔍 Enrichment Service]
    end

    subgraph UC_Auth["🔑 Authentication"]
        UC1[Login]
        UC2[Register]
        UC3[Reset Password]
        UC4[Verify Email]
    end

    subgraph UC_LeadCatch["🔗 LeadCatch Module"]
        UC5[Capture Lead]
        UC6[Parse LinkedIn Profile]
        UC7[Import CSV]
        UC8[Enrich Lead]
        UC9[Score Lead]
        UC10[Detect Duplicates]
        UC11[Export Leads]
        UC12[Categorize Lead]
    end

    subgraph UC_Appointments["📅 Appointments Module"]
        UC13[Schedule Appointment]
        UC14[Manage Availability]
        UC15[Sync Calendar]
        UC16[Book Meeting]
        UC17[Reschedule Appointment]
        UC18[Cancel Appointment]
        UC19[Send Reminders]
    end

    subgraph UC_CRM["📁 CRM Core"]
        UC20[Manage Contacts]
        UC21[Manage Companies]
        UC22[Manage Deals]
        UC23[Create Tasks]
        UC24[View Pipeline]
        UC25[View Analytics]
        UC26[Add Notes]
    end

    subgraph UC_Admin["⚙️ Administration"]
        UC27[Manage Users]
        UC28[Configure Settings]
        UC29[View Reports]
        UC30[Manage Event Types]
    end

    %% User connections
    U --> UC1
    U --> UC2
    U --> UC5
    U --> UC6
    U --> UC7
    U --> UC11
    U --> UC13
    U --> UC14
    U --> UC16
    U --> UC17
    U --> UC18
    U --> UC20
    U --> UC21
    U --> UC22
    U --> UC23
    U --> UC24
    U --> UC25
    U --> UC26

    %% Admin connections
```

```
A --> UC1
A --> UC27
A --> UC28
A --> UC29
A --> UC30

%% AI System connections
AI --> UC8
AI --> UC9
AI --> UC10
AI --> UC12
AI --> UC19

%% External System connections
CS --> UC15
CS --> UC19
ES --> UC8

%% Include relationships
UC5 -->|includes| UC8
UC5 -->|includes| UC9
UC6 -->|extends| UC5
UC7 -->|extends| UC5
UC8 -->|includes| UC12
UC16 -->|includes| UC19
UC13 -->|includes| UC15
```



Narrative Use Cases

Use Case 1: Capture and Enrich Lead

Use Case ID	UC-001
Use Case Name	Capture and Enrich Lead
Actor(s)	User (Sales Rep), AI System, Enrichment Service
Description	User captures a lead from various sources, and the system automatically enriches it with additional data using AI
Preconditions	User is logged into the system
Postconditions	Lead is stored in database with enriched data and AI-generated score

Main Flow:

1. User accesses the LeadCatch module
2. User selects input method (LinkedIn URL, Website, Manual Entry, or CSV Import)
3. User provides lead information
4. System validates input data
5. AI parses and extracts lead information
6. System initiates waterfall enrichment:
 - [Apollo.io](#) lookup
 - Clearbit lookup
 - Additional providers as needed
7. AI scores the lead (1-100)
8. AI categorizes and tags the lead
9. System checks for duplicates
10. System saves the lead to database
11. System displays enriched lead profile to user

Alternative Flows:

- **4a.** Invalid input: System displays error message, returns to step 3
- **6a.** Enrichment service unavailable: System saves partial data, flags for retry
- **9a.** Duplicate found: System prompts user to merge or create new

Use Case 2: Schedule Appointment

Use Case ID	UC-002
Use Case Name	Schedule Appointment
Actor(s)	User, External Attendee, Calendar Service, AI System
Description	User or external attendee schedules a meeting through the appointment system
Preconditions	User has configured availability and event types
Postconditions	Appointment is created and synced to all calendars

Main Flow:

1. Attendee accesses public booking page
2. Attendee selects event type
3. System retrieves user's availability from connected calendars
4. System calculates available time slots
5. Attendee selects preferred date and time
6. Attendee enters contact information
7. AI suggests optimal meeting time (optional)
8. System validates availability (no conflicts)
9. System creates appointment in database
10. System syncs to Google Calendar/Outlook
11. System sends confirmation email to both parties
12. System schedules reminder notifications

Alternative Flows:

- **5a.** No available slots: System suggests alternative dates
 - **8a.** Conflict detected: System returns to step 5 with updated availability
 - **10a.** Calendar sync fails: System queues for retry, notifies user
-

Use Case 3: Manage Sales Pipeline

Use Case ID	UC-003
Use Case Name	Manage Sales Pipeline
Actor(s)	User (Sales Rep), Admin
Description	User manages deals through different stages of the sales pipeline
Preconditions	User is logged in, deals exist in the system
Postconditions	Deal status is updated, activities are logged

Main Flow:

1. User accesses Pipeline view (Kanban board)
2. System displays deals organized by stage
3. User drags deal card to new stage
4. System updates deal status
5. System logs activity in timeline
6. System triggers automated actions (notifications, tasks)
7. System updates analytics

Alternative Flows:

- 3a. User clicks on deal: System opens deal detail view
 - 3b. User adds new deal: System opens deal creation form
-

Use Case 4: Import Leads from CSV

Use Case ID	UC-004
Use Case Name	Import Leads from CSV
Actor(s)	User, AI System
Description	User bulk imports leads from a CSV file
Preconditions	User has a CSV file with lead data
Postconditions	All valid leads are imported and enriched

Main Flow:

1. User accesses Import function in LeadCatch
2. User uploads CSV file
3. System validates file format
4. System displays field mapping interface
5. User maps CSV columns to lead fields
6. System validates data in each row
7. System detects duplicates
8. User confirms import (with duplicate handling options)
9. System imports leads in batches
10. System queues leads for AI enrichment
11. System displays import summary report

Alternative Flows:

- 3a. Invalid format: System displays supported formats, returns to step 2
 - 6a. Validation errors: System highlights invalid rows, allows correction
 - 7a. Duplicates found: System displays duplicate count, offers merge/skip options
-

Use Case 5: Generate AI-Powered Email

Use Case ID	UC-005
Use Case Name	Generate AI-Powered Email
Actor(s)	User, AI System
Description	User generates personalized email content using AI
Preconditions	User has selected a lead or contact
Postconditions	Email is generated and ready to send

Main Flow:

1. User selects a lead/contact
2. User clicks "Generate Email"
3. User selects email type (intro, follow-up, proposal, etc.)
4. System retrieves lead context (company, role, interactions)
5. AI generates personalized email draft
6. System displays draft to user
7. User reviews and edits
8. User sends or saves draft

Alternative Flows:

- **5a.** AI generation fails: System displays error, offers retry
 - **7a.** User regenerates: Returns to step 5 with modifications
-



Entity Relationship Diagram (ERD)

erDiagram

```
USERS {  
    uuid id PK  
    string email UK  
    string password_hash  
    string full_name  
    string avatar_url  
    string role  
    string timezone  
    boolean email_verified  
    timestamp created_at  
    timestamp updated_at  
}
```

```
LEADS {  
    uuid id PK  
    uuid user_id FK  
    string email UK  
    string first_name  
    string last_name  
    string phone  
    string company_name  
    string job_title  
    string linkedin_url  
    string website  
    string source  
    int score  
    string status  
    json enrichment_data  
    json custom_fields  
    text notes  
    timestamp created_at  
    timestamp updated_at  
}
```

```
CONTACTS {  
    uuid id PK  
    uuid user_id FK  
    uuid company_id FK  
    string email UK  
    string first_name  
    string last_name  
    string phone  
    string job_title  
    string linkedin_url  
    text address  
    json custom_fields  
    timestamp created_at  
    timestamp updated_at  
}
```

```
COMPANIES {  
    uuid id PK  
    uuid user_id FK  
    string name  
    string domain UK  
    string industry  
    string size  
    string linkedin_url  
    text address  
    string phone  
    json enrichment_data  
    timestamp created_at  
    timestamp updated_at  
}
```

```
DEALS {  
    uuid id PK  
    uuid user_id FK  
    uuid contact_id FK  
    uuid company_id FK  
    string name  
    decimal amount  
    string currency  
    string stage  
}
```

```
    int probability
    date expected_close_date
    text description
    timestamp created_at
    timestamp updated_at
    timestamp closed_at
}
```

```
APPOINTMENTS {
    uuid id PK
    uuid user_id FK
    uuid event_type_id FK
    uuid contact_id FK
    string title
    text description
    timestamp start_time
    timestamp end_time
    string timezone
    string status
    string location
    string meeting_url
    string google_event_id
    string outlook_event_id
    json attendees
    timestamp created_at
    timestamp updated_at
}
```

```
EVENT_TYPES {
    uuid id PK
    uuid user_id FK
    string name
    string slug UK
    text description
    int duration_minutes
    string color
    boolean is_active
    json availability_rules
    json buffer_times
    timestamp created_at
    timestamp updated_at
}
```

```
AVAILABILITY {
    uuid id PK
    uuid user_id FK
    int day_of_week
    time start_time
    time end_time
    boolean is_available
    timestamp created_at
}
```

```
CALENDAR_CONNECTIONS {
    uuid id PK
    uuid user_id FK
    string provider
    string access_token
    string refresh_token
    string calendar_id
    timestamp token_expires_at
    boolean is_active
    timestamp created_at
    timestamp updated_at
}
```

```
TASKS {
    uuid id PK
    uuid user_id FK
    uuid contact_id FK
    uuid deal_id FK
    string title
    text description
    string priority
    string status
    timestamp due_date
    timestamp completed_at
    timestamp created_at
    timestamp updated_at
}
```

```

ACTIVITIES {
    uuid id PK
    uuid user_id FK
    uuid contact_id FK
    uuid lead_id FK
    uuid deal_id FK
    string type
    string title
    text description
    json metadata
    timestamp created_at
}

NOTES {
    uuid id PK
    uuid user_id FK
    uuid contact_id FK
    uuid lead_id FK
    uuid deal_id FK
    text content
    boolean is_pinned
    timestamp created_at
    timestamp updated_at
}

TAGS {
    uuid id PK
    uuid user_id FK
    string name
    string color
    string entity_type
    timestamp created_at
}

LEAD_TAGS {
    uuid lead_id FK
    uuid tag_id FK
}

CONTACT_TAGS {
    uuid contact_id FK
    uuid tag_id FK
}

AI_LOGS {
    uuid id PK
    uuid user_id FK
    string action_type
    string model_used
    int tokens_used
    json input_data
    json output_data
    int response_time_ms
    timestamp created_at
}

%% Relationships
USERS ||--o{ LEADS : "owns"
USERS ||--o{ CONTACTS : "owns"
USERS ||--o{ COMPANIES : "owns"
USERS ||--o{ DEALS : "owns"
USERS ||--o{ APPOINTMENTS : "hosts"
USERS ||--o{ EVENT_TYPES : "creates"
USERS ||--o{ AVAILABILITY : "sets"
USERS ||--o{ CALENDAR_CONNECTIONS : "connects"
USERS ||--o{ TASKS : "creates"
USERS ||--o{ ACTIVITIES : "performs"
USERS ||--o{ NOTES : "writes"
USERS ||--o{ TAGS : "creates"
USERS ||--o{ AI_LOGS : "generates"

COMPANIES ||--o{ CONTACTS : "employs"
COMPANIES ||--o{ DEALS : "associated"

CONTACTS ||--o{ DEALS : "involved"
CONTACTS ||--o{ APPOINTMENTS : "attends"
CONTACTS ||--o{ TASKS : "related"
CONTACTS ||--o{ ACTIVITIES : "related"
CONTACTS ||--o{ NOTES : "about"

```

```
LEADS ||--o{ ACTIVITIES : "related"
LEADS ||--o{ NOTES : "about"
LEADS ||--o{ LEAD_TAGS : "has"

CONTACTS ||--o{ CONTACT_TAGS : "has"

TAGS ||--o{ LEAD_TAGS : "applied"
TAGS ||--o{ CONTACT_TAGS : "applied"

EVENT_TYPES ||--o{ APPOINTMENTS : "defines"

DEALS ||--o{ TASKS : "related"
DEALS ||--o{ ACTIVITIES : "related"
DEALS ||--o{ NOTES : "about"
```

Sequence Diagrams

Sequence Diagram 1: Lead Capture and Enrichment

```
sequenceDiagram
    autonumber
    participant U as User
    participant UI as Web Interface
    participant API as API Server
    participant AI as AI Module
    participant ENR as Enrichment Services
    participant DB as Database

    U->>UI: Enter LinkedIn URL
    UI->>API: POST /api/leads/parse
    API->>AI: parseLinkedInProfile(url)
    AI->>AI: Extract profile data
    AI-->>API: Parsed lead data

    API->>API: Initiate waterfall enrichment
    API->>ENR: Apollo.io lookup
    ENR-->>API: Company & contact data

    alt Apollo data incomplete
        API->>ENR: Clearbit lookup
        ENR-->>API: Additional data
    end

    API->>AI: scoreLead(enrichedData)
    AI->>AI: Calculate lead score
    AI-->>API: Score (0-100)

    API->>AI: categorizeLead(enrichedData)
    AI->>AI: Generate tags
    AI-->>API: Category & tags

    API->>DB: Check duplicates
    DB-->>API: Duplicate status

    API->>DB: INSERT lead
    DB-->>API: Lead created

    API->>DB: INSERT activity log
    DB-->>API: Activity logged

    API-->>UI: Lead response
    UI-->>U: Display enriched lead
```

Sequence Diagram 2: Appointment Booking

```
sequenceDiagram
    autonumber
    participant A as Attendee
    participant UI as Booking Page
    participant API as API Server
    participant CAL as Calendar Service
    participant DB as Database
    participant NOTIF as Notification Service

    A->>UI: Access booking page
    UI->>API: GET /api/event-types/{slug}
    API->>DB: Get event type details
    DB-->>API: Event type data
    API-->>UI: Event type info

    UI->>API: GET /api/availability/slots
    API->>DB: Get user availability rules
    DB-->>API: Availability config

    API->>CAL: Fetch busy times
    CAL-->>API: Busy slots

    API->>API: Calculate available slots
    API-->>UI: Available time slots
    UI-->>A: Display calendar

    A->>UI: Select time slot
    A->>UI: Enter contact info
    UI->>API: POST /api/appointments/book

    API->>DB: Verify slot still available
    DB-->>API: Confirmed

    API->>DB: INSERT appointment
    DB-->>API: Appointment created

    par Sync Calendars
        API->>CAL: Create Google event
        CAL-->>API: Event ID
        API->>CAL: Create Outlook event
        CAL-->>API: Event ID
    and Send Notifications
        API->>NOTIF: Send confirmation (Host)
        API->>NOTIF: Send confirmation (Attendee)
    end

    API->>DB: UPDATE appointment with event IDs
    API->>DB: Schedule reminder job

    API-->>UI: Booking confirmation
    UI-->>A: Success page
```

Sequence Diagram 3: AI Email Generation

```
sequenceDiagram
    autonumber
    participant U as User
    participant UI as Web Interface
    participant API as API Server
    participant AI as AI Module
    participant DB as Database

    U->>UI: Select lead/contact
    U->>UI: Click "Generate Email"
    UI->>API: GET /api/contacts/{id}
    API->>DB: Fetch contact details
    DB-->>API: Contact data

    API->>DB: Fetch interaction history
    DB-->>API: Activities & notes

    API-->>UI: Contact context

    U->>UI: Select email type
    UI->>API: POST /api/ai/generate-email

    API->>AI: Prepare prompt with context
    AI->>AI: Generate personalized email
    AI-->>API: Email draft

    API->>DB: Log AI usage
    DB-->>API: Logged

    API-->>UI: Email draft
    UI-->>U: Display draft

    U->>UI: Edit and send
    UI->>API: POST /api/email/send
    API->>DB: Log activity
    API-->>UI: Email sent
    UI-->>U: Confirmation
```



Data Flow Diagrams (DFD)

Level 0: Context Diagram

```
flowchart TB
    subgraph External["External Entities"]
        U((User))
        ATT((Attendee))
        GCAL((Google Calendar))
        OUTLOOK((Outlook))
        APOLLO((Apollo.io))
        CLEARBIT((Clearbit))
        EMAIL((Email Service))
    end

    subgraph System["0. AI-Powered CRM System"]
        CRM[AI-Powered CRM Platform]
    end

    %% User flows
    U -->|Login credentials| CRM
    U -->|Lead data, queries| CRM
    U -->|Appointment requests| CRM
    U -->|Contact/Deal updates| CRM
    CRM -->|Dashboard views| U
    CRM -->|Reports & analytics| U
    CRM -->|AI insights| U

    %% Attendee flows
    ATT -->|Booking request| CRM
    CRM -->|Available slots| ATT
    CRM -->|Confirmation| ATT

    %% Calendar integrations
    CRM <-->|Calendar sync| GCAL
    CRM <-->|Calendar sync| OUTLOOK

    %% Enrichment services
    CRM -->|Enrichment request| APOLLO
    APOLLO -->|Company/Contact data| CRM
    CRM -->|Enrichment request| CLEARBIT
    CLEARBIT -->|Enrichment data| CRM

    %% Notifications
    CRM -->|Emails & reminders| EMAIL
```

Level 1: Main Processes

```
flowchart TB
    subgraph External["External Entities"]
        U((User))
        ATT((Attendee))
        GCAL((Google Calendar))
        OUTLOOK((Outlook))
        ENRICH((Enrichment APIs))
        EMAIL((Email Service))
    end

    subgraph Processes["Processes"]
        P1[1.0 Authentication & Authorization]
        P2[2.0 Lead Management]
        P3[3.0 Appointment Scheduling]
        P4[4.0 Contact & Company Management]
        P5[5.0 Deal Management]
        P6[6.0 AI Processing]
        P7[7.0 Reporting & Analytics]
    end

    subgraph DataStores["Data Stores"]
        D1[(D1: Users)]
        D2[(D2: Leads)]
        D3[(D3: Contacts)]
        D4[(D4: Companies)]
        D5[(D5: Appointments)]
        D6[(D6: Deals)]
        D7[(D7: Activities)]
        D8[(D8: AI Logs)]
    end

    %% Authentication flows
    U -->|Credentials| P1
    P1 -->|Session token| U
    P1 <-->|User data| D1

    %% Lead Management flows
    U -->|Lead input| P2
    P2 -->|Lead list| U
    P2 -->|Enrichment request| ENRICH
    ENRICH -->|Enriched data| P2
    P2 <-->|Lead data| D2
    P2 -->|AI request| P6
    P6 -->|Score & tags| P2
    P2 -->|Activity| D7

    %% Appointment flows
    U -->|Availability config| P3
    ATT -->|Booking request| P3
    P3 -->|Available slots| ATT
    P3 -->|Confirmation| ATT
    P3 <-->|Appointment data| D5
    P3 <-->|Calendar events| GCAL
    P3 <-->|Calendar events| OUTLOOK
    P3 -->|Notifications| EMAIL
    P3 -->|Activity| D7

    %% Contact & Company flows
    U -->|Contact/Company data| P4
    P4 -->|Contact/Company list| U
    P4 <-->|Contact data| D3
    P4 <-->|Company data| D4
    P4 -->|Activity| D7

    %% Deal Management flows
    U -->|Deal updates| P5
    P5 -->|Pipeline view| U
    P5 <-->|Deal data| D6
    P5 -->|Activity| D7

    %% AI Processing flows
    U -->|AI requests| P6
    P6 -->|AI responses| U
    P6 <-->|AI logs| D8

    %% Reporting flows
```

```
U -->|Report requests| P7
P7 -->|Reports & dashboards| U
P7 <--|Activity data| D7
P7 <--|Lead analytics| D2
P7 <--|Deal analytics| D6
```

Level 2: Lead Management Process (2.0)

```
flowchart TB
    subgraph External["External"]
        U((User))
        APOLLO((Apollo.io))
        CLEARBIT((Clearbit))
        AI((AI Module))
    end

    subgraph Process2["2.0 Lead Management"]
        P2_1[2.1 Capture Lead]
        P2_2[2.2 Parse Lead Data]
        P2_3[2.3 Enrich Lead]
        P2_4[2.4 Score Lead]
        P2_5[2.5 Detect Duplicates]
        P2_6[2.6 Categorize Lead]
        P2_7[2.7 Import CSV]
        P2_8[2.8 Export Leads]
    end

    subgraph DataStores["Data Stores"]
        D2[(D2: Leads)]
        D7[(D7: Activities)]
        D9[(D9: Tags)]
    end

    %% Lead Capture
    U -->|LinkedIn URL / Website / Manual| P2_1
    P2_1 -->|Raw input| P2_2
    P2_2 -->|Parse request| AI
    AI -->|Parsed data| P2_2
    P2_2 -->|Parsed lead| P2_3

    %% Lead Enrichment (Waterfall)
    P2_3 -->|Company lookup| APOLLO
    APOLLO -->|Company data| P2_3
    P2_3 -->|Contact lookup| CLEARBIT
    CLEARBIT -->|Contact data| P2_3

    %% Lead Scoring
    P2_3 -->|Enriched lead| P2_4
    P2_4 -->|Score request| AI
    AI -->|Lead score| P2_4

    %% Duplicate Detection
    P2_4 -->|Lead for check| P2_5
    P2_5 <-->|Existing leads| D2

    %% Categorization
    P2_5 -->|Unique lead| P2_6
    P2_6 -->|Categorize request| AI
    AI -->|Categories & tags| P2_6
    P2_6 <-->|Tags| D9
    P2_6 -->|Final lead| D2
    P2_6 -->|Activity log| D7

    %% CSV Import
    U -->|CSV file| P2_7
    P2_7 -->|Validated rows| P2_1
    P2_7 -->|Import report| U

    %% Export
    U -->|Export request| P2_8
    D2 -->|Lead data| P2_8
    P2_8 -->|CSV/Excel file| U

    %% User Output
    D2 -->|Lead list| U
    P2_5 -->|Duplicate alert| U
```

Level 2: Appointment Scheduling Process (3.0)

```
flowchart TB
    subgraph External["External"]
        U((User))
        ATT((Attendee))
        GCAL((Google Calendar))
        OUTLOOK((Outlook))
        EMAIL((Email Service))
        SMS((SMS Service))
    end

    subgraph Process3["3.0 Appointment Scheduling"]
        P3_1[3.1 Configure Availability]
        P3_2[3.2 Manage Event Types]
        P3_3[3.3 Calculate Available Slots]
        P3_4[3.4 Process Booking]
        P3_5[3.5 Sync Calendars]
        P3_6[3.6 Send Notifications]
        P3_7[3.7 Reschedule/Cancel]
        P3_8[3.8 Schedule Reminders]
    end

    subgraph DataStores["Data Stores"]
        D5[(D5: Appointments)]
        D10[(D10: Event Types)]
        D11[(D11: Availability)]
        D12[(D12: Calendar Connections)]
        D7[(D7: Activities)]
    end

    %% User configures availability
    U -->|Availability rules| P3_1
    P3_1 <-->|Availability data| D11
    P3_1 -->|Confirmation| U

    %% Event type management
    U -->|Event type config| P3_2
    P3_2 <-->|Event types| D10
    P3_2 -->|Event type list| U

    %% Slot calculation
    ATT -->|Date range request| P3_3
    P3_3 <-->|Availability rules| D11
    P3_3 <-->|Event type settings| D10
    P3_3 -->|Fetch busy times| P3_5
    P3_5 <-->|Busy times| GCAL
    P3_5 <-->|Busy times| OUTLOOK
    P3_5 -->|Busy slots| P3_3
    P3_3 -->|Available slots| ATT

    %% Booking process
    ATT -->|Booking request| P3_4
    P3_4 <-->|Check conflicts| D5
    P3_4 -->|Create appointment| D5
    P3_4 -->|Sync request| P3_5
    P3_5 -->|Create event| GCAL
    P3_5 -->|Create event| OUTLOOK
    P3_5 -->|Event IDs| P3_4
    P3_4 -->|Activity log| D7

    %% Notifications
    P3_4 -->|Send confirmations| P3_6
    P3_6 -->|Email| EMAIL
    P3_6 -->|SMS| SMS
    P3_6 -->|Confirmation| ATT
    P3_6 -->|Notification| U

    %% Reminders
    P3_4 -->|Schedule reminders| P3_8
    P3_8 -->|Reminder triggers| P3_6

    %% Reschedule/Cancel
    U -->|Reschedule/Cancel request| P3_7
    ATT -->|Reschedule/Cancel request| P3_7
    P3_7 <-->|Update appointment| D5
```

```
P3_7 -->|Update calendar| P3_5
P3_7 -->|Notify| P3_6
```

Level 2: AI Processing (6.0)

```
flowchart TB
    subgraph External["External"]
        U((User))
        OPENAI((OpenAI GPT-4))
    end

    subgraph Process6["6.0 AI Processing"]
        P6_1[6.1 Lead Parsing]
        P6_2[6.2 Lead Scoring]
        P6_3[6.3 Lead Categorization]
        P6_4[6.4 Email Generation]
        P6_5[6.5 Smart Scheduling]
        P6_6[6.6 Token Management]
    end

    subgraph DataStores["Data Stores"]
        D8[(D8: AI Logs)]
        D2[(D2: Leads)]
        D3[(D3: Contacts)]
    end

    %% Lead Parsing
    U -->|Raw text/URL| P6_1
    P6_1 -->|Prompt| OPENAI
    OPENAI -->|Extracted data| P6_1
    P6_1 -->|Parsed lead| U
    P6_1 -->|Log| D8

    %% Lead Scoring
    U -->|Lead data| P6_2
    D2 -->|Lead context| P6_2
    P6_2 -->|Scoring prompt| OPENAI
    OPENAI -->|Score| P6_2
    P6_2 -->|Lead score| U
    P6_2 -->|Log| D8

    %% Categorization
    U -->|Lead for tagging| P6_3
    P6_3 -->|Categorization prompt| OPENAI
    OPENAI -->|Tags & category| P6_3
    P6_3 -->|Categorized lead| U
    P6_3 -->|Log| D8

    %% Email Generation
    U -->|Email request| P6_4
    D3 -->|Contact context| P6_4
    P6_4 -->|Email prompt| OPENAI
    OPENAI -->|Email draft| P6_4
    P6_4 -->|Generated email| U
    P6_4 -->|Log| D8

    %% Smart Scheduling
    U -->|Scheduling query| P6_5
    P6_5 -->|Scheduling prompt| OPENAI
    OPENAI -->|Time suggestions| P6_5
    P6_5 -->|Suggested times| U
    P6_5 -->|Log| D8

    %% Token Management
    P6_1 -->|Token count| P6_6
    P6_2 -->|Token count| P6_6
    P6_3 -->|Token count| P6_6
    P6_4 -->|Token count| P6_6
    P6_5 -->|Token count| P6_6
    P6_6 -->|Usage tracking| D8
    P6_6 -->|Rate limiting| OPENAI
```

Technology Stack Summary

Layer	Technology
Frontend	Next.js 14+, React 18+, TypeScript, Tailwind CSS, Shadcn UI
Backend	Next.js API Routes, Drizzle ORM
Database	PostgreSQL (Supabase)
Authentication	Supabase Auth
AI	OpenAI GPT-4o
Calendar	Google Calendar API, Microsoft Outlook API
Enrichment	Apollo.io , Clearbit
Notifications	Resend/SendGrid (Email), Twilio (SMS)
Deployment	Vercel

References

- Next.js Documentation: <https://nextjs.org/docs>
- Supabase Documentation: <https://supabase.com/docs>
- OpenAI API Reference: <https://platform.openai.com/docs>
- Drizzle ORM Documentation: <https://orm.drizzle.team/docs>
- Google Calendar API: <https://developers.google.com/calendar>
- Microsoft Graph API: <https://docs.microsoft.com/en-us/graph>

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Prepared by:

- Moaz Hassan Ismail El Garawany (320240032)
- Ahmed Hossam Aldeen Alsayed (320240051)
- Ahmed Mahmoud (320240024)